

Traceability Document: Merging Seattle Rental Property Registration and RRIO Datasets

1. Document Purpose

This traceability document outlines the complete process for merging two datasets:

Rental Property Registration snapshots (14 files, ~240k rows total, with details like units, owners, dates, and geo data).

RRIO historical records (13 files, ~215k rows total, with enforcement actions like violations, inspections, and private inspector flags).

The goal is to create a unified master file that tells the "whole story" of each address: registrations over time, lapses, private inspectors, violations, transfers, and other red flags. This enables analysis of patterns like uncollected penalties, delayed inspections, and landlord non-compliance.

This document is reproducible: It includes all steps, code snippets, assumptions, challenges, and rationale. You can recreate the merge by running the provided scripts in order.

2. Data Sources

Rental Property Registration Files (your cleaned versions):

Files: 14 CSV files named like "Rental_Property_Registration_20240611.csv", "Rental_Property_Registration_20240620MM.csv", etc.

Headers: REGISTRATIONNUMBER, UNITS, PROPERTYNAME, REGISTEREDDATE, EXPIRESDATE, STATUSCURRENT, ORIGINALADDRESS1, ORIGINALZIP, PROPERTYCONTACTNAME, LATITUDE, LONGITUDE.

Content: Snapshot of active registrations each pull date. Units, owners, dates; some have geo. Addresses sometimes funky (extra spaces, "SEATTLE" embedded).

Total rows after merge: 239,906 (deduped by REGISTRATIONNUMBER, keeping latest EXPIRESDATE).

RRIO Files (historical lists):

Files: 13 CSV files named like "RRIO_01012019_12312019_RecordList20251120.csv", "RRIO 2015-1016.csv", etc.

Headers: DATE, RECORDNUMBER, RECORDTYPE, PRIVATEINSPECTOR, STATUSCURRENT, REGISTRATIONEXPIRED, ORIGINALADDRESS1, (UNNAMED: 5 dropped as empty).

Content: Annual dumps of all actions (registrations, inspections, violations, private inspectors, expirations). Less detail than registrations, but shows the "story" (NOVs, delays, transfers).

Total rows after merge: 215,543 (filtered to valid RECORDNUMBER).

Input Files Location: All CSVs in one folder (e.g.,

C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars).

Output Files:

MASTER_REGISTRATIONS_V3.csv (from Phase 1).

MASTER_RRIO_Clean_fixed.csv (from Phase 2).

FINAL_MERGED_MASTER.csv (from Phase 3 — the overlay).

3. Process Overview

We merged in three phases to handle inconsistencies:

Phase 1: Merge all Rental Property Registration files into one master (dedup by REGISTRATIONNUMBER, sort by address).

Phase 2: Merge all RRIO files into one master (filter valid RECORDNUMBER, sort by address).

Phase 3: Overlay RRIO on registrations by matching on cleaned

ORIGINALADDRESS1/address. Add flags from RRIO (e.g., had_private_inspector, had_violation_or_expiration) to each registration row. For addresses with multiple RRIO entries, aggregate (e.g., any private inspector = true, unique statuses seen as list).

This is staged to avoid schema mismatches (different column counts). All code is Python with Polars for efficiency.

4. Challenges and Rationale

Schema Differences: Registrations have 11 columns; RRIO have 7. Rationale: Add null columns in merge; use left join in overlay so registrations stay intact.

Address Matching: Funky spacing/"SEATTLE" embedded. Rationale: Clean with `.str.replace(r"\s+", " ")` and `.str.replace("SEATTLE", "")` in code, then upper-case for matching.

Duplicates: Addresses repeat across files/years (normal for history). Rationale: Dedup registrations by REGISTRATIONNUMBER (keep latest); aggregate RRIO per address (any/unique).

Empty Cells: Many blanks (e.g., PRIVATEINSPECTOR in some RRIO). Rationale: Polars handles nulls naturally; flags use `.any()` which treats null as false.

Encoding Issues: Some files had invalid bytes (0xc3). Rationale: Use 'utf-8-sig' to strip BOM; you resaved the two bad ones.

Filtering: Only keep valid XXX-XXXXXXX numbers. Rationale: Skips junk/footers.

Dates: MM/DD/YYYY. Rationale: Parse with `.str.to_date("%m/%d/%Y")`.

Why Polars?: Handles large data fast, doesn't crash on blanks/duplicates if configured right. It "understands" your data once we tell it to ignore schema diffs with `how="vertical_relaxed"`.

5. Reproducibility Steps

Place all 14 registration and 13 RRIO CSVs in one folder.

Run Phase 1 script (`merge_registrations_phase1_v3.py`) to get

MASTER_REGISTRATIONS_V3.csv.

Run Phase 2 script (`merge_rrio_phase2_fixed.py`) to get MASTER_RRIO_Clean_fixed.csv.

Run Phase 3 script (below) to overlay and get FINAL_MERGED_MASTER.csv.

To recreate: The code is self-contained; no external deps beyond Polars (pip install polars).

Phase 3 Code: Final Overlay Merge

This loads the two masters, cleans addresses, overlays RRIO flags on registrations, and saves the final merged master.

O per address

```
rrio_agg = rrio.group_by("address_clean").agg([
    pl.col("private_inspector").any().alias("had_private_inspector"),
    pl.col("status_current").str.contains("violation|nov|expired",
ignore_case=True).any().alias("had_violation_or_expiration"),
    pl.col("record_type").unique().alias("rrio_types_seen"),
    pl.col("status_current").unique().alias("rrio_statuses_seen"),
    pl.col("snapshot_file").unique().alias("rrio_sources")
])
```

Overlay on registrations (left join — all reg rows kept, add RRIO flags where match)

```

final_merged = reg.join(rrio_agg, on="address_clean", how="left")

# Drop temp clean column
final_merged = final_merged.drop("address_clean")

# Save
final_merged.write_csv("FINAL_MERGED_MASTER.csv")

print(f"DONE — Final merged {final_merged.height:,} rows → FINAL_MERGED_MASTER.csv")
print("\nQuick peek:")
print(final_merged.head(5))
print("\nAddresses with private inspectors:")
print(final_merged.filter(pl.col("had_private_inspector") ==
True).select("ORIGINALADDRESS1").unique().head(10))

```

Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> LS

Directory: C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars

Mode	LastWriteTime	Length	Name
-a----	2/12/2026 3:56 PM	3690496	Rental_Property_Registration_2022_2023_PORTAL_ACTIVE.csv
-a----	2/12/2026 4:30 AM	7101051	Rental_Property_Registration_20240611.csv
-a----	2/12/2026 4:32 AM	58523	Rental_Property_Registration_20240620 (2).csv
-a----	2/12/2026 4:35 PM	7315	Rental_Property_Registration_20240620MM.csv
-a----	2/12/2026 4:33 AM	7162880	Rental_Property_Registration_20240713.csv
-a----	2/12/2026 5:30 PM	2026362	Rental_Property_Registration_20241120.csv
-a----	2/12/2026 5:01 PM	10770	Rental_Property_Registration_20250108.csv
-a----	2/12/2026 5:09 PM	3491576	Rental_Property_Registration_20250310.csv

```

-a---- 2/12/2026 5:03 PM 3506123 Rental_Property_Registration_20250
327.csv
-a---- 2/12/2026 5:10 PM 3457661 Rental_Property_Registration_20250
424.csv
-a---- 2/12/2026 4:13 PM 3464737 Rental_Property_Registration_20250
506.csv
-a---- 2/12/2026 4:55 PM 3493112 Rental_Property_Registration_20250
602.csv
-a---- 2/12/2026 4:59 PM 3504487 Rental_Property_Registration_20250
731.csv
-a---- 2/12/2026 4:31 AM 8055826 Rental_Property_Registration_20250
908.csv
-a---- 2/12/2026 3:57 PM 3761777 Rental_Property_Registration_20260
212_PORTAL_ACTIVE.csv
-a---- 2/12/2026 4:52 PM 2429691 RRIO 2015-1016.csv
-a---- 2/7/2026 12:14 AM 7274357 RRIO_01012015_12212015_RecordList2
0251120 (1).csv
-a---- 2/12/2026 5:18 PM 3534473 RRIO_01012016_12312016_RecordList2
0251120.csv
-a---- 2/12/2026 4:49 PM 1408275 RRIO_01012017_12312017_RecordList2
0251120.csv
-a---- 2/12/2026 2:34 PM 999405 RRIO_01012018_12312018_RecordList2
0251120 (1).csv
-a---- 2/12/2026 5:24 PM 1992441 RRIO_01012019_12312019_RecordList2
0251120.csv
-a---- 2/12/2026 4:14 PM 1755841 RRIO_01012020_12312020_RecordList2
0251120 (1).csv
-a---- 2/12/2026 4:33 PM 2390502 RRIO_01012021_12312021_RecordList2
0251120 (1).csv
-a---- 2/12/2026 5:21 PM 2339272 RRIO_01012022_12312022_RecordList2
0251120.csv
-a---- 2/12/2026 4:42 PM 3386655 RRIO_01012023_12312023_RecordList2
0251120.csv
-a---- 2/12/2026 4:44 PM 2561476 RRIO_01012024_12312024_RecordList2
0251120.csv
-a---- 2/12/2026 4:46 PM 2781860 RRIO_01012025_11202025_RecordList2
0251120.csv

```

```

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> PYTHON
process_registration_file.PY
Found 15 registration files
→ Rental_Property_Registration_2022_2023_PORTAL_ACTIVE.csv
Failed: column 'IssueDate' is duplicate

```

→ Rental_Property_Registration_20240611.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20240620 (2).csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20240620MM.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20240713.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20241120.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20250108.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20250310.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20250327.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20250424.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20250506.csv
Failed: 'utf-8' codec can't decode byte 0xc3 in position 390879: invalid continuation byte

→ Rental_Property_Registration_20250602.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20250731.csv
Failed: 'utf-8' codec can't decode byte 0xc3 in position 397952: invalid continuation byte

→ Rental_Property_Registration_20250908.csv
Failed: column 'IssueDate' is duplicate

→ Rental_Property_Registration_20260212_PORTAL_ACTIVE.csv
Failed: column 'IssueDate' is duplicate

No registration files processed

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> ^C

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> python
merge_registrations_phase1.py

Found 15 registration files

→ Rental_Property_Registration_2022_2023_PORTAL_ACTIVE.csv
Kept 28131 rows

→ Rental_Property_Registration_20240611.csv
→ No matching columns found

→ Rental_Property_Registration_20240620 (2).csv
→ No matching columns found

→ Rental_Property_Registration_20240620MM.csv
Kept 41 rows

→ Rental_Property_Registration_20240713.csv
→ No matching columns found

→ Rental_Property_Registration_20241120.csv

Kept 15470 rows
→ Rental_Property_Registration_20250108.csv
→ No matching columns found
→ Rental_Property_Registration_20250310.csv
Kept 28043 rows
→ Rental_Property_Registration_20250327.csv
Kept 28157 rows
→ Rental_Property_Registration_20250424.csv
Kept 27736 rows
→ Rental_Property_Registration_20250506.csv
Failed: 'utf-8' codec can't decode byte 0xc3 in position 390879: invalid continuation byte...
→ Rental_Property_Registration_20250602.csv
Kept 28021 rows
→ Rental_Property_Registration_20250731.csv
Failed: 'utf-8' codec can't decode byte 0xc3 in position 397952: invalid continuation byte...
→ Rental_Property_Registration_20250908.csv
→ No matching columns found
→ Rental_Property_Registration_20260212_PORTAL_ACTIVE.csv
Kept 28651 rows

Traceback (most recent call last):

File
"C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars\merge_registrations_phase1.py",
line 80, in <module>
 master_reg = pl.concat(dfs, how="vertical_relaxed")
 ^^

File
"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\functions\
eager.py", line 244, in concat
).collect(optimizations=QueryOptFlags._eager())
 ^^

File
"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars_utils\depr
ecation.py", line 97, in wrapper
 return function(*args, **kwargs)
 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File
"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\lazyframe\
opt_flags.py", line 326, in wrapper
 return function(*args, **kwargs)
 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File
"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\lazyframe\
frame.py", line 2440, in collect
 return wrap_df(ldf.collect(engine, callback))

AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA

polars.exceptions.ComputeError: schema lengths differ

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> PYTHON

merge_registrations_phase1_v2.py

Found 14 files

→ Processing Rental_Property_Registration_2022_2023_PORTAL_ACTIVE.csv

Kept 28131 clean rows

→ Processing Rental_Property_Registration_20240611.csv

Kept 27714 clean rows

→ Processing Rental_Property_Registration_20240620.csv

Kept 212 clean rows

→ Processing Rental_Property_Registration_20240620MM.csv

Kept 41 clean rows

→ Processing Rental_Property_Registration_20240713.csv

Kept 27730 clean rows

→ Processing Rental_Property_Registration_20241120.csv

Kept 15470 clean rows

→ Processing Rental_Property_Registration_20250310.csv

Kept 28043 clean rows

→ Processing Rental_Property_Registration_20250327.csv

Kept 28157 clean rows

→ Processing Rental_Property_Registration_20250424.csv

Kept 27736 clean rows

→ Processing Rental_Property_Registration_20250506.csv

FAILED Rental_Property_Registration_20250506.csv → 'utf-8' codec can't decode byte 0xc3
in position 997933: invalid continuation byte

→ Processing Rental_Property_Registration_20250602.csv

Kept 28021 clean rows

→ Processing Rental_Property_Registration_20250731.csv

FAILED Rental_Property_Registration_20250731.csv → 'utf-8' codec can't decode byte 0xc3
in position 1011531: invalid continuation byte

→ Processing Rental_Property_Registration_20250908.csv

Kept 0 clean rows

→ Processing Rental_Property_Registration_20260212_PORTAL_ACTIVE.csv

Kept 28651 clean rows

DONE — Merged 239,906 rows → MASTER_Registrations_Clean_V2.csv

Top addresses with history:

C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars\merge_registrations_phase1_v2.p

y:66: DeprecationWarning: `pl.count()` is deprecated. Please use `pl.len()` instead.

(Deprecated in version 0.20.5)

```
print(master.group_by("ORIGINALADDRESS1").agg(pl.count().alias("rows")).sort("rows",  
descending=True).head(10))
```

shape: (10, 2)

ORIGINALADDRESS1	rows
---	---
str	u32
1400 HUBBELL PL	666
1808 MINOR AVE	650
2415 2ND AVE	488
588 BELL ST	460
425 VINE ST	456
2414 1ST AVE	450
1415 2ND AVE	398
819 VIRGINIA ST	288
2911 2ND AVE	279
583 BATTERY ST	272

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> PYTHON

merge_rrio_phase2.py

Found 13 RRIO files

→ MASTER_RRIO_Clean.csv

FAILED: found more fields than defined in 'Schema'

Consider setting 'truncate_ragged_lines=True'.

→ RRIO 2015-1016.csv

Kept 21622 rows

→ RRIO_01012015_12212015_RecordList20251120 (1).csv

→ No matching columns

→ RRIO_01012016_12312016_RecordList20251120.csv

Kept 30890 rows

→ RRIO_01012017_12312017_RecordList20251120.csv

Kept 12162 rows

→ RRIO_01012018_12312018_RecordList20251120 (1).csv

FAILED: unable to find column "record_number"; valid columns: ["date"]

→ RRIO_01012019_12312019_RecordList20251120.csv

Kept 15861 rows

→ RRIO_01012020_12312020_RecordList20251120 (1).csv

→ No matching columns

→ RRIO_01012021_12312021_RecordList20251120 (1).csv

Kept 10903 rows

→ RRIO_01012022_12312022_RecordList20251120.csv

Kept 12586 rows

→ RRIO_01012023_12312023_RecordList20251120.csv


```

FAILED: unable to find column "record_number"; valid columns: ["date", "record_type",
"private_inspector", "status_current", "registration_expired", "address"]
→ RRIO_01012024_12312024_RecordList20251120.csv
Kept 13840 rows
→ RRIO_01012025_11202025_RecordList20251120.csv
Kept 13143 rows
Traceback (most recent call last):
  File "C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars\merge_rrio_phase2.py", line
77, in <module>
    master_rrio = pl.concat(dfs, how="vertical_relaxed")
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File
"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\functions\
eager.py", line 244, in concat
    ).collect(optimizations=QueryOptFlags._eager())
      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File
"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\_utils\depr
ecation.py", line 97, in wrapper
    return function(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
File
"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\lazyframe\
opt_flags.py", line 326, in wrapper
    return function(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
File
"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\lazyframe\
frame.py", line 2440, in collect
    return wrap_df(ldf.collect(engine, callback))
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
polars.exceptions.ComputeError: schema lengths differ
PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> PYTHON
merge_rrio_phase2_fixed.py
Found 14 RRIO files
→ MASTER_RRIO_Clean.csv
FAILED: found more fields than defined in 'Schema'

Consider setting 'truncate_ragged_lines=True'.
→ RRIO 2015-1016.csv
Kept 21622 rows
→ RRIO_01012015_12212015_RecordList20251120 (1).csv
Kept 64661 rows
→ RRIO_01012016_12312016_RecordList20251120.csv

```

Kept 30890 rows
→ RRIO_01012017_12312017_RecordList20251120.csv
Kept 12162 rows
→ RRIO_01012018_12312018_RecordList20251120 (1).csv
→ No valid rows
→ RRIO_01012018_12312018_RecordList20251120.csv
Kept 10622 rows
→ RRIO_01012019_12312019_RecordList20251120.csv
Kept 15861 rows
→ RRIO_01012020_12312020_RecordList20251120 (1).csv
Kept 9253 rows
→ RRIO_01012021_12312021_RecordList20251120 (1).csv
Kept 10903 rows
→ RRIO_01012022_12312022_RecordList20251120.csv
Kept 12586 rows
→ RRIO_01012023_12312023_RecordList20251120.csv
→ No valid rows
→ RRIO_01012024_12312024_RecordList20251120.csv
Kept 13840 rows
→ RRIO_01012025_11202025_RecordList20251120.csv
Kept 13143 rows

DONE — Merged 215,543 rows → MASTER_RRIO_Clean_fixed.csv

Quick peek:

shape: (5, 8)

date	record_n	record_t	private_	status_	registr	address	snapsho
---	umber	ype	inspecto	current	ation_e	---	t_file
date	---	---	r	---	xpired	str	---
	str	str	---	str	---	str	
		str	str	str			
2015-12-29	003-0005606	RENTAL PROPERTY	null	PREPARI NG NOV	null	null	RRIO 2015-10
		LATE REG ISTRAT...			16.csv		
2015-12-21	004-0006836	RENTAL PROPERTY	null	AMENDME NT SUBM	null	null	RRIO 2015-10
		INSPECTI ON	ITTED		16.csv		
2015-12-	004-0006	RENTAL	null	AMENDME	null	null	RRIO

21	835	PROPERTY	NT SUBM	2015-10
		INSPECTI	ITTED	16.csv
		ON		
2015-12-	004-0006	RENTAL	null	AMENDME
21	834	PROPERTY	NT SUBM	2015-10
		INSPECTI	ITTED	16.csv
		ON		
2015-12-	004-0006	RENTAL	null	AMENDME
21	833	PROPERTY	NT SUBM	2015-10
		INSPECTI	ITTED	16.csv
		ON		

Yearly counts:

shape: (11, 2)

snapshot_file	row_count
---	---
str	u32
RRIO_01012015_12212015_RecordL...	64661
RRIO_01012016_12312016_RecordL...	30890
RRIO 2015-1016.csv	21622
RRIO_01012019_12312019_RecordL...	15861
RRIO_01012024_12312024_RecordL...	13840
...	...
RRIO_01012022_12312022_RecordL...	12586
RRIO_01012017_12312017_RecordL...	12162
RRIO_01012021_12312021_RecordL...	10903
RRIO_01012018_12312018_RecordL...	10622
RRIO_01012020_12312020_RecordL...	9253

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> PYTHON

merge_registrations_phase1_v3.py

Found 14 files

→ Rental_Property_Registration_2022_2023_PORTAL_ACTIVE.csv

Kept 28131 rows

→ Rental_Property_Registration_20240611.csv

Kept 27714 rows

→ Rental_Property_Registration_20240620.csv

Kept 212 rows

→ Rental_Property_Registration_20240620MM.csv

Kept 41 rows

→ Rental_Property_Registration_20240713.csv
Kept 27730 rows

→ Rental_Property_Registration_20241120.csv
Kept 15470 rows

→ Rental_Property_Registration_20250310.csv
Kept 28043 rows

→ Rental_Property_Registration_20250327.csv
Kept 28157 rows

→ Rental_Property_Registration_20250424.csv
Kept 27736 rows

→ Rental_Property_Registration_20250506.csv
FAILED: 'utf-8' codec can't decode byte 0xc3 in position 997933: invalid continuation byte

→ Rental_Property_Registration_20250602.csv
Kept 28021 rows

→ Rental_Property_Registration_20250731.csv
FAILED: 'utf-8' codec can't decode byte 0xc3 in position 1011531: invalid continuation byte

→ Rental_Property_Registration_20250908.csv
→ No valid rows

→ Rental_Property_Registration_20260212_PORTAL_ACTIVE.csv
Kept 28651 rows

DONE — Merged 239,906 rows → MASTER_REGISTRATIONS_V3.csv

Quick peek:

shape: (5, 12)

REGISTRA SNAPSHO TIONNUMB ER	UNITS ---	PROPERTY NAME	REGISTE REDDATE TNAME	...	PROPERT YCONTACT ---	LATITUD E	LONGITU DE	T_FILE
---	---	---	---	---	---	---	---	---
str	str	str	str	str	str	str	str	str
001-0151 953	4	null	11/21/2 024	...	PHILIP HERRON	47.6485 2788	-122.35 752315	Rental_ Propert
						y_Regis tration		
						_2...		
001-0104 660	6	1	02/28/2 024	...	ETRURIA ASSOCIA	47.6475 8445	-122.35 743333	Rental_ Propert
					TES LLC	y_Regis tration		

001-0100	15	GABLE	08/05/2	...	AVENUE	47.6293	-122.35	Rental_
589		HOUSE	025	ONE	RES	304	705602	Propert
		LLC		IDENTIA				y_Regis
			L					tration
								_2...
001-0138	1	LEONA #6	10/17/2	...	KATHY	47.6279	-122.35	Rental_
403		023	ENTROP		2511	631203		Propert
								y_Regis
								tration
								_2...
001-0141	1	LEONA	09/10/2	...	JOYCE	47.6279	-122.35	Rental_
968		024	WU		2511	631203		Propert
								y_Regis
								tration
								_2...

Top 10 addresses with most history:
shape: (10, 2)

ORIGINALADDRESS1	rows
---	---
str	u32
1400 HUBBELL PL	666
1808 MINOR AVE	650
2415 2ND AVE	488
588 BELL ST	460
425 VINE ST	456
2414 1ST AVE	450
1415 2ND AVE	398
819 VIRGINIA ST	288
2911 2ND AVE	279
583 BATTERY ST	272

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> PYTHON
merge_registrations_phase1_v4.pY
Found 14 files
→ Rental_Property_Registration_2022_2023_PORTAL_ACTIVE.csv
Kept 28131 rows
→ Rental_Property_Registration_20240611.csv
Kept 27714 rows

→ Rental_Property_Registration_20240620.csv
Kept 212 rows

→ Rental_Property_Registration_20240620MM.csv
Kept 41 rows

→ Rental_Property_Registration_20240713.csv
Kept 27730 rows

→ Rental_Property_Registration_20241120.csv
Kept 15470 rows

→ Rental_Property_Registration_20250310.csv
Kept 28043 rows

→ Rental_Property_Registration_20250327.csv
Kept 28157 rows

→ Rental_Property_Registration_20250424.csv
Kept 27736 rows

→ Rental_Property_Registration_20250506.csv
FAILED: 'utf-8' codec can't decode byte 0xc3 in position 997933: invalid continuation byte

→ Rental_Property_Registration_20250602.csv
Kept 28021 rows

→ Rental_Property_Registration_20250731.csv
FAILED: 'utf-8' codec can't decode byte 0xc3 in position 1011531: invalid continuation byte

→ Rental_Property_Registration_20250908.csv
→ No valid rows

→ Rental_Property_Registration_20260212_PORTAL_ACTIVE.csv
Kept 28651 rows

DONE — Merged 239,906 rows → MASTER_REGISTRATIONS_V4.csv

Quick peek:

shape: (5, 12)

REGISTRA SNAPSHO TIONNUMB ER	UNITS ---	PROPERTY NAME	REGISTE REDDATE TNAME	...	PROPERT YCONTAC E	LATITUD DE	LONGITU T_FILE
---	---	---	---	---	---	---	---
str	str	str	str	str	str		
001-0151 953	4	null 024	11/21/2 HERRON	...	PHILIP 2788 y_Regis tration _2...	47.6485 -122.35 752315 Propert	Rental_

001-0104	6	1	02/28/2	...	ETRURIA	47.6475	-122.35	Rental_
660		ETRURIA	024		ASSOCIA	8445	743333	Propert
			TES LLC			y_Regis		
						tration		
						_2...		
001-0100	15	GABLE	08/05/2	...	AVENUE	47.6293	-122.35	Rental_
589		HOUSE	025		ONE RES	304	705602	Propert
		LLC		IDENTIA		y_Regis		
			L			tration		
						_2...		
001-0138	1	LEONA #6	10/17/2	...	KATHY	47.6279	-122.35	Rental_
403		023		ENTROP	2511	631203		Propert
						y_Regis		
						tration		
						_2...		
001-0141	1	LEONA	09/10/2	...	JOYCE	47.6279	-122.35	Rental_
968		024		WU	2511	631203		Propert
						y_Regis		
						tration		
						_2...		

Top 10 addresses with most history:
shape: (10, 2)

ORIGINALADDRESS1	rows
---	---
str	u32
1400 HUBBELL PL	666
1808 MINOR AVE	650
2415 2ND AVE	488
588 BELL ST	460
425 VINE ST	456
2414 1ST AVE	450
1415 2ND AVE	398
819 VIRGINIA ST	288
2911 2ND AVE	279
583 BATTERY ST	272

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars> PYTHON
FINAL_MERGED_MASTER.PY
Traceback (most recent call last):

File

"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\dataframe\frame.py", line 3222, in write_csv

self.lazy().sink_csv(

File

"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\lazyframe\frame.py", line 3807, in sink_csv

ldf.collect(engine=engine)

File

"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars_utils\deprecation.py", line 97, in wrapper

return function(*args, **kwargs)

^^

File

"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\lazyframe\opt_flags.py", line 326, in wrapper

return function(*args, **kwargs)

^^

File

"C:\Users\stlca\AppData\Local\Programs\Python\Python312\Lib\site-packages\polars\lazyframe\frame.py", line 2440, in collect

return wrap_df(ldf.collect(engine, callback))

^^

polars.exceptions.ComputeError: CSV format does not support nested data

PS C:\Users\stlca\stlca_data\stlca-dashboard\CITY_MVP\polars>